Five-anomaly model evaluation

Data, testing and iterative improvement • completed studies through 19 October 2026

SmolVLM learns useful task rankings. Camera transfer and false alarms remain unresolved.

6 + 3

Changed-weight runs

6 CLIP visual + 3 SmolVLM LoRA

3,792

SmolVLM held-out cases

10 source/task cohorts; retrospective

70.8%

Unattended endpoint FPR

94.6% recall at dev-budget cutoff

Next decisive evidence: independently reviewed event histories and long normal streams on new cameras.

Five objectives require different evidence

Existing dataset annotations and definitions remain the source of truth

Objective	Evidence required	Current qualification
Smoking	Person + hand/object/action evidence	Low-resolution CCTV differs from photos
Littering	Disposal action and scene context	IWDD includes larger waste dumping
Loitering	Identified person and timed history	A visible person alone is insufficient
Rough sleeping	Persistent apparent public sleep	Posture photos are provisional proxies
Unattended items	Item persistence + owner separation	30 s criterion belongs to AVSS source

Unknown history or missing labels stay unknown; other objectives are never automatically labelled normal.

Adaptation now includes SmolVLM language weights

Five requested model families • protocols and trainable components remain explicit

Model	Completed adaptation / evaluation	Current comparison scope
MobileCLIP2-S2	Visual fine-tuning; region / temporal heads	Five targets; actual visual updates
SigLIP2-B/16-224	Visual fine-tuning; region / temporal heads	Five targets; actual visual updates
SmolVLM-500M	NEW: language-layer LoRA; binary margins	Five targets; matched original control
MiniCPM-V 4.0	Deterministic prompted image responses	Historical expanded image comparison
VadCLIP	Separate supervised offline smoking adapter	Different split / supervision; unchanged

Three SmolVLM adapters were trained. Its frozen vision tower differs from the earlier CLIP visual updates.

From common prompts to controlled model adaptation

Counts describe separate studies and reuse some sources

Stage	Completed work	Evidence / qualification
13–14 Oct	13,400 then 55,156 predictions	Four-model comparison; qualified labels
15–16 Oct	6 actual CLIP visual fine-tunes	2 encoders × 3 seeds; changed weights
16 Oct	330 configurations; 50 video heads	Crops, source transfer, tracks and events
17 Oct	38 policies; state replay; 830 views	Ensembles, memory and fixed tiling
18 Oct	83-page full technical report	Documentation; no new training that day
19 Oct	3 SmolVLM LoRA runs; 10 cohorts	Seed replication; threshold failures retained

The updated report and deck preserve earlier versions. No new inference-speed or latency test was added.

Many files did not provide complete event labels

The dataset audit changed what could be evaluated honestly

Usable supervised evidence

Smoking: CCTV person/action labels
IWDD: dumping action clips + controls
Street Scene: person track/type labels
CAVIAR: item/person source boxes
ABODA: reviewed time intervals

Useful diagnostics, weaker event truth

6,667 littering candidate images
1,675 person/loitering candidates
Synthetic lying / falling postures
Static public-sleep photographs
MEVA abandonment-action windows

Positive-only images, augmented copies and pose labels cannot establish five-event precision or false-alarm rates.

Training covered all five targets with uneven evidence

First supervised study: 7,498 examples • 12,478 distinct view images

Objective	Train (normal / +)	Dev (normal / +)	Test (normal / +)
Smoking, CCTV + train photos	1,381 / 716	722 / 413	253 / 393
Littering / dumping	130 / 144	29 / 32	50 / 50
Loitering	282 / 41	85 / 13	64 / 11
Public sleep	49 / 25	14 / 7	12 / 6
Unattended items	416 / 74	137 / 181	720 / 242

A separate 806-photo smoking holdout is reported separately. Endpoints and adjacent frames remain correlated.

More data was acquired and screened for compatibility

Source diversity, duplicate control and label meaning matter more than download size

Source	Acquired / retained	Role and limit
Smoking photos + HMDB	1,118 photos; 906 clips	Photo learning; action-source transfer
Thermal loitering	874,936 observations; 31 dates	Provided tracks; one thermal camera
MEVA + public sleep	13 videos; 28 new-author photos	Action proxy; provisional photo labels
AVSS2007	3 videos; 2 clean primary events	One conservative source block
17 Oct author release	3 new clips + 3 ABODA aliases	30 highway-debris component frames

The latest release adds small-object stress cases, but adds no verified owner-associated unattended events.

Keep related sources together and freeze decisions

A source holdout is more informative than a random split of adjacent frames

Before evaluation

Preserve source / camera / author groups
Respect official video/photo partitions
Remove exact-pixel aliases
Purge known shared-film part aliases
Fit routes and thresholds on dev only

Limits that remain

Previously inspected tests are retrospective
Movie / actor identities are incomplete
Only a few physical cameras / events
Provisional interval / sleep labels
Repeated seeds add no new domains

No test-optimal threshold or lucky seed is substituted after seeing results.

Accuracy runs used available GPU compute

No pacing, CPU emulation or new latency benchmarks in the later studies

Quality execution

RTX 5000 Ada GPU
Batched CLIP extraction / head fitting
Deterministic VLM generation
Failures and abstentions stay counted
Pinned weights, inputs and raw scores

Numerical consistency

Fine-tuning: BF16 autocast, F32 weights
Final CLIP scoring: F32, TF32 off
Batch-shape sensitivity was audited
New detector: 830 F32 GPU views
Event delay measures task detection

Event onset delay is a quality metric; it is separate from computation latency.

Ranking, frame decisions and alerts are different

All three are needed to judge practical detection

Image / clip discrimination

AP and AUROC: score ranking
Recall + FPR: a fixed operating point
F1 + confusion counts: class balance
VLM hard labels: no invented AP
Weak candidates: prediction rate only

Continuous event detection

Past-only 20-second histories
Actual alert emission times
One matched alarm per labelled event
Normal false, repeated, unknown alerts
Onset delay + reviewed normal hours

An alert in an unreviewed transition is unknown; an early continuing alarm is not silently rematched to a later onset.

Expanded unattended images did not yield a clear winner

962 test endpoints • 242 positive / 720 normal • two camera families

Model	Precision	Recall	F1	FPR
MobileCLIP2-S2	25.3%	75.2%	0.379	74.6%
SigLIP2-B/16-224	0.0%	0.0%	0.000	2.6%
SmolVLM-500M	18.6%	63.6%	0.288	93.8%
MiniCPM-V 4.0	77.1%	34.7%	0.479	3.5%

MiniCPM has the best raw frame F1 here, while missing 65% of positive endpoints. Labels remain provisional.

Both CLIP visual backbones were updated

Two encoders × seeds 42 / 43 / 44 • frozen early layers and text towers

What learned

MobileCLIP: 18.1M visual parameters
SigLIP: 14.2M visual parameters
Last vision stage + five binary heads
Masked target losses and balancing
Scene-preserving augmentation

How overfitting was constrained

Pretrained feature / weight penalties
Frozen early layers and batch norms
Probe and pretrained-score controls
Development-selected routes / epochs
Three-seed variation retained

These six visual fine-tunes are distinct from later frozen-feature classifiers, GRUs and ensembles.

Fine-tuning helped some tasks and hurt others

Same-study pretrained score → development-selected primary route (AP)

Evidence	Cases	MobileCLIP AP	SigLIP AP
Smoking CCTV camera c1	646	.642 → .564	.669 → .650
Waste dumping clips	100	.424 → .858	.499 → .885
Loitering windows	75 (11 +)	.320 → .093	.097 → .097
Public-sleep photos	18 (6 +)	.502 → .976	.733 → .915
Unattended endpoints	962	.209 → .652	.309 → .896

Photo smoking improves strongly (.901 / .954 AP), but photo success did not transfer to CCTV.

The next screen combined data, crops and adaptation

33 configurations per target / backbone • five objectives • two CLIP encoders

Inputs and representations

Train: 3,258 → 3,973 examples
Automatic contextual person/item crops
Full / region visual representations
Temporal pooling vs ordered views
Sparse detection geometry

Candidate routes

Linear and nonlinear classifiers
Pretrained / adapted weight blends
Alpha: 0, .25, .5 and 1
Dev-source AP route selection
Raw, dev-F1 and FPR-budget cutoffs

330 configurations include references and 320 classifier fits; they are not 330 separate backbone fine-tunes.

Historical AP rose unevenly after the strategy screen

Previous selected bundle → new selected strategy • same previously inspected cases

Objective	MobileCLIP AP	SigLIP AP	Evidence limitation
Smoking CCTV	.564 → .690	.650 → .722	One held-out camera
Waste dumping	.858 → .806	.885 → .893	Full-clip action classification
Loitering	.093 → .489	.097 → .550	One Street Scene camera
Public sleep	.976 → .931	.915 → .883	Small provisional photo set
Unattended	.652 → .980	.896 → .925	Provisional overlapping endpoints

MobileCLIP dumping and both public-sleep photo routes regress. Higher AP still does not solve alert thresholds.

The broader transfer tests exposed remaining failures

Previous bundle → selected strategy AP • before later HMDB adaptation

Transfer evidence	Cases	MobileCLIP AP	SigLIP AP
New smoking-photo holdout	224	.953 → .997	.995 → 1.000
HMDB movie/web action clips	906	.605 → .576	.685 → .664
New-author public-sleep photos	28	.925 → .952	.943 → .933
MEVA abandonment-action proxy	27	.569 → .545	.637 → .621

Photo-domain gains coexist with movie/web and MEVA regressions. These sources are not interchangeable.

Loitering and public sleep remain data-limited

Their useful component results do not yet validate robust event detection

Loitering component

17,878 eligible supplied trajectories
Chronological 19 / 6 / 6-date split
Forest test AP .577 vs duration .283
Development AP .863
One thermal camera; no RGB tracking

Public-sleep component

28 new-author photo cases
MobileCLIP AP .952; SigLIP .933
Provisional visible-sleep labels
No confirmed persistence / duration
Rest, lying and bedding need controls

The next model must see the evidence the target requires: complete tracks or persistent state, with matched negatives.

Small people and missing items were concrete bottlenecks

117 deliberately selected errors were provisionally reviewed across the five targets

Smoking spatial evidence

Median smoker height: 11.2 px at 224
71.8% of positive frames below 12 px
Old crop misses half every target: 40.2%
Multi-person / upper crops: 0.3%
Coverage is separate from recognition

Item and ownership evidence

CAVIAR item recall at IoU .5:
Faster R-CNN tracks: 0 / 136
Grounding DINO tracks: 66 / 136
Proximity can confuse passers-by
Jitter can reset stationarity

Error-board findings are triage, not an independently reviewed estimate of failure causes.

Separate appearance from tracked event state

Automatic proposals → frozen CLIP features → causal heads → explicit alerts

1. Detect

Full scene + up to four persons
Whole / upper-body crops
Two item / owner contexts

2. Represent

Frozen CLIP visual features
Causal geometry and timestamps
Linear / GRU / instance heads

3. Decide

Past-only scores / same-item state
Persistence and alert emission
Event / normal / unknown counts

No annotation boxes choose inference crops. Missing owner history never becomes an invented ownership relation.

Fifty heads tested region, sequence and person evidence

8,931 modeling examples after the known-film source guard

Smoking: 14 heads / encoder

Scene and region linear baselines
Region GRU, three seeds
Region + geometry GRU, three seeds
Person-supervised MIL, three seeds
Hard-negative variant, three seeds

Unattended: 11 heads / encoder

Scene and region linear baselines
Region GRU, three seeds
COCO-geometry GRU, three seeds
Grounding-geometry GRU, three seeds
Unchanged visual CLIP features

Route and epoch selection maximize worst development-source AP, then mean AP. All seeds and regressions remain visible.

CCTV gains did not repeat consistently across backbones

205 HMDB evaluation clips now follow 530 grouped training clips; CCTV remains static

Model / source	Previous bundle	Pilot scene	Selected pilot
MobileCLIP / CCTV 646	.690	.626	.725
SigLIP / CCTV 646	.722	.717	.688
MobileCLIP / HMDB 205	.697	.842	.908
SigLIP / HMDB 205	.762	.888	.778

SigLIP’s development-selected pilot route regresses on CCTV and trails its own scene baseline on HMDB.

Ensembling improves HMDB; low-FPR CCTV remains hard

38 dev-only policies • every dev source must meet the 5% empirical FPR constraint

Policy	CCTV AP	Recall / FPR	HMDB AP	Recall / FPR
MobileCLIP / previous	0.725	16.5% / 7.9%	0.908	86.7% / 2.9%
MobileCLIP / 17 Oct	0.725	10.9% / 5.5%	0.908	86.7% / 2.3%
SigLIP / previous	0.688	2.3% / 1.6%	0.778	66.7% / 1.7%
SigLIP / 17 Oct	0.720	1.8% / 1.6%	0.964	76.7% / 1.7%

SigLIP selects the all-head ensemble. MobileCLIP retains the same person head with a stricter cutoff.

Explicit item state was more useful than a global score

Two clean AVSS events • 2.76 normal minutes • first used 16 Oct, now retrospective

Route / policy	Events	Normal false / repeats	Median onset delay
MobileCLIP / global score	2 / 2	1 / 3	25.8 s
SigLIP / global score	2 / 2	6 / 3	12.1 s
MobileCLIP + known state gate	2 / 2	0 / 0	36.3 s
SigLIP + known state gate	2 / 2	0 / 1	34.8 s
Known source state rule alone	2 / 2	0 / 0	34.8 s

The rule requires the same stationary item for 30 s and its previously associated owner absent for ≥0.5 s.

A good AP or zero observed alarms is not a guarantee

Insufficient normal footage is a measurable limitation

Historical calibration diagnostic

Entire normal recordings held aside
Both arms exclude those recordings
SigLIP: 0/4 → 1/4 events
Normal false alerts: 4 → 1
Still 26.3 false alerts / normal hour

Uncertainty in the AVSS result

2 events and 165.74 normal seconds
0 observed state-rule false alerts
95% one-sided Poisson upper bound:
65.1 false alerts / normal hour
Camera transfer remains untested

Even under a homogeneous Poisson assumption, about 3 zero-alert normal hours are needed for an upper bound below 1/hour.

More state machinery did not improve the AVSS result

Original detections and 30-second source rule are held fixed

Ablation	Events	Normal false / repeats	Median onset delay
Original global state debounce	2/2	0 / 0	34.82 s
Original tracks + item memory	2/2	0 / 0	34.82 s
Median stationarity + global	2/2	0 / 0	35.07 s
Median stationarity + memory	2/2	0 / 0	35.07 s

Keep the simpler state baseline: smoothing adds 0.25 s; item memory increases repeats on the censored HARD diagnostic.

The new highway clips are component stress tests

Pinned author release: three bundled clips, three explicit ABODA aliases

Acquisition and preparation

28.6 MB author archive
Three bundled highway videos
7.6–14.6 seconds each
Ten annotated positive frames / video
Actual FPS and timestamps checked

What cannot be inferred

Debris boxes do not establish an owner
No 30-second stationary evidence
No complete reviewed normal intervals
ABODA links add no independent data
No new unattended-event accuracy

The additional frames test detection of very small objects; they are not newly verified unattended events.

Fixed overlapping tiles test the small-object bottleneck

166 source-annotated frames × five views = 830 new GPU image views

Reference source	IoU	Full image recall	Full + four tiles recall
CAVIAR item source boxes	0.3	69/136 (50.7%)	117/136 (86.0%)
CAVIAR item source boxes	0.5	66/136 (48.5%)	116/136 (85.3%)
Highway-debris stress cases	0.3	0/30 (0.0%)	7/30 (23.3%)
Highway-debris stress cases	0.5	0/30 (0.0%)	7/30 (23.3%)

Same weights, query and score thresholds. Partial reference boxes support recall, not object precision or event accuracy.

VadCLIP remains a qualified smoking baseline

Offline supervised adaptation • different source groups and supervision

Measured test evidence

1,056 test frames
560 positive / 496 normal
AP 0.633; AUROC 0.587
Recall 51.8%; FPR 35.7%
F1 0.565, dev-frozen cutoff

Comparison boundaries

9 training clips / 4 validation clips
Adapter receives smoking supervision
Not an unsupervised five-task model
No new VadCLIP fine-tune this round
Not the same split as CLIP camera c1

Avoid a single leaderboard that mixes this offline supervised protocol with image-model zero-shot or event-stream tests.

Use the original model as a matched weight control

One task query per example • real first-answer scores • same inputs before / after adaptation

Matched within this study

Same accepted target definitions
Same 512-pixel letterbox images
Same ordered views and visual tokens
Same binary Yes / No task query
Same first-token logit margin

Comparison limits

Earlier study generated six-label JSON
Prompt / preprocessing also changed
No learned unknown or explanations
No boxes or tracked identities added
Retrospective source holdouts

Attribute adaptation gains to the matched original-to-LoRA comparison; preserve the old generation results.

All five tasks are trained; evidence is uneven

3,973 train • 1,812 development • 2,025 test • 806 historical external photos

Objective	Train normal / +	Dev normal / +	Test normal / +
Smoking	1739 / 1073	812 / 502	365 / 505
Littering	130 / 144	29 / 32	50 / 50
Loitering	282 / 41	85 / 13	64 / 11
Public sleep	49 / 25	14 / 7	12 / 6
Unattended	416 / 74	137 / 181	720 / 242

8,616 main examples use 13,596 distinct images. Unknown objectives are masked; source groups stay together.

Ten cohorts cover 3,792 unique held-out examples

Main holdouts plus 961 additional transfer cases • no transfer labels enter training or selection

Additional transfer

HMDB: 906 clips, 109 smoking
New-author sleep: 28 photos, 17 +
MEVA: 27 windows, 15 actions
7,375 extra distinct decoded images
20,971 total decoded images checked

What the data can establish

HMDB: movie / web action transfer
Sleep: provisional visible context
MEVA: action, not persistent state
Exact pixels / groups checked
Old holdouts remain retrospective

More frames and seeds do not add independent cameras. This round reuses previously collected data.

Adapt 4.34 million language parameters

32 language layers × 7 projections • rank 8 • alpha 16 • dropout 0.05

What changes

q/k/v/o + gate/up/down projections
224 adapted language modules
4,341,760 trainable parameters
Answer-only full-vocabulary loss
~17 MB saved weights per adapter

Frozen inputs and optimization

Vision + connector + embeddings frozen
64 cached visual tokens / image
1 static / 4 temporal / 8 clip views
BF16 training; F32 final scoring
AdamW; effective batch 32

This is model-weight adaptation. It does not update the image encoder or learn tracking and ownership.

Three trained candidates all select epoch one

Selection: mean source AP within each task, then equal macro average over five tasks

Seed	Learning rate	Best epoch	Epochs run	Dev macro AP
42	3e-05	1	3	0.6584
42	0.0001	1	3	0.7026
43	0.0001	1	3	0.6548

Original dev AP 0.5657 → primary joint 0.7026. Task routing uses the low LR only for public sleep.

Training loss falls while development AP declines

Primary high learning rate: 7,946 example visits per epoch • selected checkpoint = epoch 1

Epoch	Train loss	Dev macro AP	Checkpoint status
1	0.4789	0.7026	SELECTED
2	0.2895	0.6485	Retained history
3	0.2235	0.6276	Retained history

Source transfer deteriorates before the planned six-epoch limit.

Early stopping protects the selected checkpoint; the lowest-loss final weights are not promoted.

LoRA improves selected rankings and regresses on loitering

Primary seed 42 • identical inputs and binary scoring • AP is ranking, not accuracy

Task / source	Total / +	Original AP	Joint AP	Routed AP
Smoking photos / Kaggle	224 / 112	0.9670	0.9844	0.9844
Smoking / CCTV	646 / 393	0.5999	0.6225	0.6225
Littering / IWDD	100 / 50	0.5922	0.7660	0.7660
Loitering / Street Scene	75 / 11	0.4250	0.1822	0.1822
Public sleep / historical	18 / 6	0.7437	0.6327	0.7885
Unattended / historical	962 / 242	0.4762	0.6933	0.6933

Joint sleep regresses; low-LR routing helps. CCTV AP 0.6225 is only slightly above constant AP 0.6084.

Photo and action-source gains do not validate events

No transfer example is used for SmolVLM training, route selection or calibration

Source	Total / +	Original AP	Joint AP	Routed AP
Smoking / Stanford40	806 / 140	0.5831	0.8900	0.8900
Smoking / HMDB	906 / 109	0.7638	0.7837	0.7837
Public sleep / new authors	28 / 17	0.8062	0.8441	0.8904
MEVA abandonment action	27 / 15	0.6076	0.6457	0.6457

All four routed APs rise. Labels remain source-specific: action clips, provisional sleep, and abandonment actions.

The development FPR cutoff fails on unattended items

Each target cutoff allows ≤5% pooled development FPR; held-out FPR can exceed it

Task / source	Recall	FPR	True / false positives
Smoking / CCTV	1.8%	0.0%	7 / 0
Littering / IWDD	30.0%	8.0%	15 / 4
Loitering / Street Scene	9.1%	6.2%	1 / 4
Public sleep / historical	33.3%	8.3%	2 / 1
Unattended / historical	94.6%	70.8%	229 / 510

Unattended: 510 / 720 normals flagged. CCTV: only 7 / 393 positives found; raw margin zero finds none.

The second seed repeats the loitering failure

Fixed primary winning learning rate • one joint adapter per seed • same held-out sources

Task / source	Seed 42 joint AP	Seed 43 joint AP
Smoking / CCTV	0.6225	0.6295
Littering / IWDD	0.7660	0.7656
Loitering / Street Scene	0.1822	0.1314
Public sleep / historical	0.6327	0.9167
Unattended / historical	0.6933	0.7052

Sleep varies greatly on six positives. Optimization replication adds no independent camera evidence.

CLIP strategies lead on most historical cohorts

Same 3,792 held-out IDs / labels; crops, resolution and training differ across pipelines

Task / source	MobileCLIP AP	SigLIP AP	Smol routed AP
Smoking / CCTV	0.6902	0.7225	0.6225
Littering / IWDD	0.8064	0.8934	0.7660
Loitering / Street Scene	0.4894	0.5505	0.1822
Public sleep / historical	0.9306	0.8833	0.7885
Unattended / historical	0.9803	0.9250	0.6933
Smoking / HMDB	0.5764	0.6639	0.7837

SmolVLM is stronger on 906-case HMDB transfer. Later CLIP video heads use a different 205-case protocol.

Require a development gain in both optimization seeds

Rule introduced after primary held-out inspection • optional, exploratory and retrospective

Target	Primary dev gain	Seed-43 dev gain	Optional route
Smoking	+0.0872	+0.0828	High LR
Littering	+0.1589	+0.1017	High LR
Loitering	+0.0308	-0.1620	Original
Rough Sleeping	+0.3084	+0.3426	Low LR
Unattended Items	+0.1253	+0.0805	High LR

Only loitering returns to the original. Primary results stay unchanged; unattended false positives remain.

Check actual checkpoints and native image execution

Saved research bundle includes adapter weights, portable base identity, task routes and cutoffs

Experiment reconstruction

7,248 dev scores replayed
15,168 held-out scores replayed
20,971 decoded images checked
3 finite changed adapters checked
20 native GPU / cached comparisons

Software and saved output

187 unit tests passed
6 real-input / missing-context checks
Max native score error < 0.000073
Portable adapters + processor files
Research bundle; no event certification

Functional CLI checks verify scores and guards. They are not six successful accuracy predictions.

No single model reliably covers all five events

Use source-qualified candidates; do not replace event evidence with a model confidence score

Objective	Useful evidence so far	Main remaining gap
Smoking	Photo learning; action ensembles; crops	CCTV low-FPR recall; tiny people
Littering	CLIP / LoRA learn IWDD dumping actions	Small disposal vs placing; online onset
Loitering	Geometry / provided tracks; original control	LoRA regression; identities / new cameras
Public sleep	Photo probes; low-LR VLM route	Tiny provisional sets; persistent video
Unattended	Grounded items + explicit state rule	Owner truth; threshold shift; normal hours

Ranking gains are real within their protocols. Reliable five-event camera transfer is still unproven.

Generalization needs evidence beyond another training run

All tested interventions remain auditable, including unsuccessful choices

Useful progress

Correct labels and source-group guards
True CLIP and language-weight updates
Dumping and photo ranking gains
Matched original controls and replays
Explicit item state and tiled recall

Limits that repeated

Photo success does not settle CCTV
Higher AP can still yield high FPR
Later epochs can overfit quickly
Seed agreement adds no new cameras
Sparse views lack owner / dwell truth

The optional LoRA safeguard is retrospective. A new frozen source holdout is needed to validate it.

Freeze new camera evidence before selecting another model

Existing definitions retained • improve source diversity, temporal labels and realistic controls

Priority	Concrete work	Acceptance evidence
1. Label / collect	New cameras; full event / owner histories	Independent review + matched controls
2. Components	Person / item coverage; identity; persistence	Automatic inputs; explicit error audit
3. Adapt	Source-balanced confirmed examples; early stop	Original control; multiple source folds
4. Validate	Freeze route / cutoff; untouched long streams	Recall, delay, FAR, repeats and bounds

Keep unknown when the required visible, duration or ownership evidence is missing.

Updated paper, editable slides and source-linked findings

Complete historical account plus detailed SmolVLM methods, results and limitations

Detailed technical paper

All original data / testing stages
New full SmolVLM training chapter
All 60 method / seed / source cells
Exact cutoffs and epoch histories
LaTeX source + portable source ZIP

Presentation and evidence

Editable PPTX + matching vector PDF
Updated one-slide summary
Speaker notes + source identities
Rendered previews + layout checks
Earlier report / slides preserved

This documentation build reads completed results. It runs no new model evaluation or latency benchmark.